

# FINITE-SAMPLE SAFE MODEL RANKING UNDER SUBGROUP-MIX TURNOVER

Anonymous authors

Paper under double-blind review

## ABSTRACT

At deployment the class/subgroup mixture of a fixed pool of candidate models can shift (label-prior / subgroup-mix turnover) while each subgroup’s class-conditional risk stays unchanged, so a model optimal for the collected mixture may cease to be optimal—and a worse model can be selected. We formalize *safe ranking* as a finite-sample certification problem: given a calibration split, decide for every future mixture weight  $w$  whether to *commit* a model with a certificate that its regret (loss minus the oracle-best model’s loss at  $w$ ) is at most an operator-set tolerance  $\tau$ , and otherwise *abstain* honestly rather than hard-select a possibly-overturning pick. We prove a linear-band structure (P1) giving joint coverage  $\geq 1 - \delta$  of every mixture risk by weighted Clopper–Pearson bands, and a tighter *paired-difference* (MCB-style) certificate (P3/Thm. 1) that bounds  $\max_j(R_{i^*} - R_j)$  by pairing models on the *same* calibration points, cancelling shared error. On four public carriers and four classifiers, certified commits always have true regret  $\leq \tau$  (coverage = 1.0), and naively *minimizing the upper band* is shown to be a vacuous or *anti-selective* selector under heteroskedastic group widths—so we use *point-estimate selection + certificate gating* (M2/M2.5), abstaining exactly on turnover-skew mixtures where forced point-estimate selection has higher observed regret. The paired certificate exposes an explicit *certificate-price frontier*: the exact finite-sample bounds commit 0.269–0.260 (matched 5-seed means); a separately labelled asymptotic normal diagnostic has a 0.503 committed rate, but is not finite-sample evidence. Under a *label budget* (Sec. 6) soundness persists (0 violations over 4200 cells); uniform spread out-commits variance-aware rules on MNIST but is beaten by them on Fashion at low budget, and a time-uniform adaptive certificate charges an adaptivity width no carrier repays. Finally, we use a fixed-budget *certificate-width surrogate* (Sec. 7) to diagnose why allocations differ. Its water-filling calculation is not an optimization theorem for the actual candidate- and mixture-indexed regret UCB or a safe allocation rule.

## 1 INTRODUCTION

Model selection is usually done once, on the mixture where data was collected. But at deployment the label/subgroup mix can move—a model trained on general web traffic now serves a skewed query stream; a classifier tuned on balanced data is deployed onto a single class-dominant population. Under *label-prior shift* (*subgroup-mix turnover*) the class-conditional distribution  $P(X \mid Y=g)$  is fixed but the mixture weight  $w$  over subgroups is not. A model  $i$ ’s risk is then linear in  $w$ ,

$$R_i(w) = \sum_g w_g r_{i,g}, \quad r_{i,g} = \mathbb{E}[\ell(\text{pred}_i, Y) \mid Y=g], \quad (1)$$

so the identity of the best model,  $\arg \min_i R_i(w)$ , can change as  $w$  moves across the simplex. Standard selection heuristics live at one of two extremes: *single-point* selection (pick by the collected-mixture estimate) ignores  $w$  entirely; *worst-group* / *DRO* / *balanced-accuracy* selection ignores where  $w$  actually is. None answers the deployment question: *for a given future mixture  $w$ , can I both pick a model and certify its regret in finite sample—and when must I abstain rather than risk a pick that overturns  $w \mapsto w'$ ?*

We study this as a *finite-sample safe ranking* problem with abstention: from a held-out calibration split, produce for each  $w$  a decision *commit*(model  $i^*$  and a certificate that its regret  $\leq \tau$ ) or *ab-*

stain (report that  $w$  is undecidable). The certificate is a finite-sample, coverage-guaranteed upper bound on the chosen model’s regret, and is allowed to abstain where the certificate is wider than the operator-set tolerance (Sec. 5, Tab. 1). Abstention is a certificate outcome, not an operational-utility claim: its fallback, latency, and monetary costs are unmeasured here.

## 1.1 CONTRIBUTIONS

1. **Linear-band ranking structure** (Sec. 3): if each entry  $r_{i,g}$  is covered by a Clopper–Pearson interval with cell confidence  $\delta/(Gk)$ , then every mixture risk satisfies  $L_i(w) \leq R_i(w) \leq U_i(w)$  with joint coverage  $\geq 1 - \delta$  (P1, Lemma 1); best-vs-rest regret of any chosen model has a *sound* upper bound  $U_{i^*}(w) - \min_j L_j(w)$  (P2).
2. **Paired-difference (MCB-style) certificate** (P3, Thm. 1): rather than subtract two *absolute* bands, bound  $\max_j (R_{i^*} - R_j)$  by *paired within-CAL* differences (same points, same group), giving a strictly tighter one-sided bound at identical joint coverage and finite-sample rigor.
3. **A mechanism-level caution on “minimize-the-upper-band” selection** (Sec. 4): this pessimistic rule degenerates to the point estimate on homogeneous widths (vacuity) and *can* reverse rank opinion under heteroskedastic widths (Prop. 3); on real data it disagrees with point-estimate selection without a consistent direction. Framed as *bands gate, they do not choose*.
4. **Certificate-price frontier** (Sec. 5, Tab. 2): the exact paired Maurer–Pontil variant commits 0.260 vs. M2’s 0.269 (matched 5-seed) with reported committed coverage 1.000. The 0.503 normal/CLT result is reported only as an asymptotic diagnostic, never as finite-sample support.
5. **Subgroup label allocation under a budget (M3, Sec. 6)**: at equal budget, uniform *out-commits* variance-aware rules on MNIST but is beaten on Fashion at low budget (carrier-heterogeneous); a time-uniform CS certificate for the adaptive track charges a width price ( $\sim 4.5\times$ ) no carrier repays, so static rules are the better-funded route (Lemma 2). Soundness holds at every budget (0 violations over 4200 cells).
6. **Fixed-budget width-surrogate diagnostic (M3.5, Sec. 7)**: for a stylized  $\beta_g/\sqrt{n_g}$  width proxy, a single fixed mixture has the familiar water-filling solution  $n_g \propto (w_g \beta_g)^{2/3}$  on positive coefficients (and zero allocation on zero coefficients). We use this only to interpret same-budget allocation experiments; it does *not* optimize the paired regret UCB, prove safe allocation, or prescribe a deployment gate.

## 2 PROBLEM AND SETUP

**Candidates.** A fixed pool of  $k$  candidate models  $\mathcal{M} = \{M_1, \dots, M_k\}$  trained on a *fit* split, disjoint from all evaluation data.

**Subgroups.** Data come from  $G$  subgroups (we use ground-truth classes). The deployment prior is a weight vector  $w$  on the simplex  $\Delta^{G-1}$ ; per-model mixture risk is  $R_i(w) = \sum_g w_g r_{i,g}$  ((1)).

**Assumption 1** (subgroup-mix turnover). *During deployment only the mixture weight  $w$  shifts; each group’s class-conditional distribution  $P(X | Y=g)$  is unchanged, so each  $r_{i,g}$  is constant. Reward evaluation uses group-IPS reweighting.*

**Calibration.** A calibration split (disjoint from fit and test) with  $n_g$  points in group  $g$ . The per-point error indicator of model  $M_i$  in group  $g$  is a Bernoulli with mean  $r_{i,g}$ . We get  $\hat{p}_{i,g}$ , a Clopper–Pearson exact interval  $[l_{i,g}, u_{i,g}]$  at cell confidence  $\delta_{\text{cell}} = \delta/(Gk)$ .

**Decision problem.** For each deployment mixture  $w$ , output either *commit*( $i^*$ , certificate “true regret  $\leq \tau$ ”) or *abstain*. Regret of a choice  $i$  at  $w$  is

$$\text{regret}(i, w) = R_i(w) - \min_j R_j(w),$$

i.e. the loss of choosing  $i$  versus the oracle-best model at  $w$ . We certify this quantity from calibration data *only*, never touching test.

### 3 THEORY: LINEAR BANDS AND THE REGRET CERTIFICATE

Throughout,  $k = |\mathcal{M}|$ ,  $G$  subgroups,  $\delta \in (0, 1)$ . Let  $\delta_{\text{cell}} = \delta/(Gk)$ , and let  $[l_{i,g}, u_{i,g}]$  be the exact CP interval for  $r_{i,g}$  at confidence  $\delta_{\text{cell}}$ .

**Lemma 1** (Linear band joint coverage). *Let  $E$  be the event that  $r_{i,g} \in [l_{i,g}, u_{i,g}]$  for all  $(i, g)$ . By Bonferroni,  $\Pr(E) \geq 1 - \delta$ . On  $E$ , for every  $w \in \Delta^{G-1}$  and every  $i$ ,*

$$L_i(w) := \sum_g w_g l_{i,g} \leq R_i(w) \leq \sum_g w_g u_{i,g} =: U_i(w).$$

*Proof.* Each  $l, u$  is a CP bound for that cell; Bonferroni over the  $Gk$  cells gives  $\Pr(E) \geq 1 - \delta$ . On  $E$ , fix  $w$ ; since  $w_g \geq 0$ ,  $\sum_g w_g = 1$ , termwise monotonicity of the (deterministic) linear combination gives the sandwich.  $\square$

**Proposition 1** (Sound minimax regret bound (P2)). *Let  $i^*$  be any chosen model and define  $\text{UB}_{i^*}(w) = U_{i^*}(w) - \min_j L_j(w)$ . On  $E$ ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{i^*}(w).$$

*Proof.*  $R_{i^*}(w) \leq U_{i^*}(w)$  (Lemma 1), and  $\min_j R_j(w) \geq \min_j L_j(w)$  since  $R_j \geq L_j$  pointwise. Subtracting the two (weaker) inequalities at their respective endpoints yields the bound; this holds for the comparator  $j$  attaining the minimum.  $\square$

Proposition 1 is *correct* but, as we show empirically (Sec. 4), unsuitable as a selector. It is a good *gate*: committing  $i^*$  whenever  $\text{UB}_{i^*}(w) \leq \tau$  certifies true regret  $\leq \tau$  with coverage  $\geq 1 - \delta$ . The tighter the bound, the more mixtures can be committed. This motivates the paired certificate, which is the central finite-sample construction.

**Definition 1** (Paired-difference certificate). *For ordered pairs  $(i, j)$ ,  $i \neq j$ , define the per-point paired difference on the calibration set*

$$d_{ij}(\mathbf{x}) = \mathbb{K}\{\text{err}_i(\mathbf{x})\} - \mathbb{K}\{\text{err}_j(\mathbf{x})\} \in \{-1, 0, +1\},$$

*and its group- $g$  mean  $\bar{d}_{ij,g}$ , an unbiased estimator of  $r_{i,g} - r_{j,g}$ . Let  $\alpha = \delta/(k(k-1)G)$  (Bonferroni over all ordered pairs  $\times$  groups). For each  $(i, j, g)$  take a one-sided upper confidence bound  $\text{UCB}_{ij,g} \geq r_{i,g} - r_{j,g}$  simultaneously with probability  $\geq 1 - \delta$  over all pairs/groups. Candidate variants (Sec. 5, Tab. 2) operate on  $X_{ij} = (\bar{d}_{ij} + 1)/2 \in [0, 1]$ . The certificate value for chosen  $i^*$  at  $w$  is*

$$\text{UB}_{\text{cf}}(i^*, w) = \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g}.$$

**Theorem 1** (Paired-difference certificate is sound (MCB-style)). *On the simultaneous event of Definition 1 (probability  $\geq 1 - \delta$ ), for any choice  $i^*$ ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{\text{cf}}(i^*, w).$$

*Hence if  $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$ , committing  $i^*$  is certified to have true regret  $\leq \tau$ , with coverage  $\geq 1 - \delta$ .*

*Proof.*  $\text{regret}(i^*, w) = \max_j (R_{i^*}(w) - R_j(w))$  because the minimum is attained by some  $j_*$  and the max over  $j$  includes  $j_*$ . Fix the worst-case  $j_*$  achieving the max. Since  $E$  holds,  $R_{i^*}(w) - R_{j_*}(w) = \sum_g w_g (r_{i^*,g} - r_{j_*,g}) \leq \sum_g w_g \text{UCB}_{i^*j_*,g}$ . This sum is exactly one term of the max in  $\text{UB}_{\text{cf}}$ , hence is bounded by it.  $\square$

The key is *paired* cancellation:  $\text{UB}_{\text{cf}}$  subtracts models evaluated on the *same* calibration points within each group, removing the shared per-point error that inflates  $\text{UB}_{i^*}$  in Provision 1.

**Honest scope of the certificate.** Each  $\text{UCB}_{ij,g}$  is computed once, independently of  $w$  (Bonferroni over ordered pairs  $\times$  groups only, Def. 1), so the simultaneous event is a single event of probability  $\geq 1 - \delta$  under which  $\text{UB}_{\text{cf}}(i^*, w) = \max_j \sum_g w_g \text{UCB}_{i^*j,g}$  is valid at the same time for *all*  $w$  on the simplex and all data-dependent choices  $i^*(w)$ : the paired certificate is *simultaneous over the continuous simplex*, with no grid-level Bonferroni. We verify this with a dense off-grid Dirichlet  $w$ -sample (Sec. 7, 1000 points per (carrier, seed, budget), none on-grid): committed-cell coverage stays 1.0 (e.g. MNIST  $\pi=0.5$  commits 2816/3000 off-grid rows). The certificate certifies *soundness* of commits, not optimal abstention;  $\tau$  is an operator parameter, and we report the coverage–commitment tradeoff as a fact, not a theorem.

## 4 WHY BANDS MUST GATE, NOT CHOOSE: M1 VERIFIED NEGATIVE

The most tempting “safety” rule is minimax selection,

$$i_{\text{mmp}}^* = \arg \min_i \left( \max_j \text{UB}_{ij}(w) \right), \quad \text{UB}_{ij}(w) = U_i(w) - L_j(w),$$

i.e. pick the model whose worst-case gap to any other is smallest. We ran it on real data (digits, Fashion-MNIST, MNIST, 20News;  $G=\text{class}$ ;  $k=4$ ; LR/C-variants, SVC, MLP).

**Proposition 2** (*M1 degeneracy*).  $\max_j (U_i(w) - L_j(w)) = U_i(w) - \min_j L_j(w)$ , so  $i_{\text{mmp}}^* = \arg \min_i U_i(w)$ : the “max joint” collapses to the minimizer of the upper band, whose lower side ( $\min_j L_j$ ) is constant across  $i$ .

When CP widths are homogeneous,  $\arg \min_i U_i(w)$  equals the point-estimate argmin (vacuity); under *heteroskedastic* widths the rule is driven by the most pessimistic band and can pick a strictly worse model. On our real grid it matches point-estimate selection on every homogeneous-width row and diverges on mixture-ball vertices without a consistent direction (App. A): the rank opinion is driven by band widths, not by the point estimate—so it neither vacuous-commits nor reliably out-ranks point-estimate selection; bands gate, they do not choose (Prop. 3).

**Proposition 3** (*Anti-selectivity of M1 under heteroskedastic widths*). *For a fixed  $w$ , suppose model  $a$  has a wide lower-side band driven by a sparse subgroup (so  $\min_j L_j(w)$  is small), and the point estimate prefers model  $b$  with  $\hat{R}_b < R_a$ . Then M1 selects  $a$  (the pessimistic pick) whenever the width  $\Delta_a = U_a - L_a$  exceeds  $\hat{R}_b - L_a$ ; committed regret becomes  $R_b - R_a$  (positive) while cal-prior commits  $b$  at regret 0 (i.e. M1 is strictly worse whenever the width of the pessimistic model exceeds the true gap).*  $\square$

*Proof.* Follows directly from Prop. 2: M1 picks  $\arg \min_i U_i(w)$ ; on the event that  $U_a(w) < U_b(w)$  while  $\hat{R}_a > \hat{R}_b$ , the rule selects the higher-error model, and the “regret” it believes it controls is an artifact of the pessimistic band width, not a real comparison.  $\square$

Safety bands are *certificates*, not arrows; we adopt the design they motivate.

## 5 POINT-ESTIMATE SELECTION WITH CERTIFICATE GATING (M2 / M2.5)

### 5.1 THE VERIFIED POLICY

Selection is driven by the point estimate, never by the bands:

$$i^*(w) = \arg \min_i \hat{R}_i(w) = \arg \min_i \sum_g w_g \hat{p}_{i,g}.$$

Bands serve *only* as a gate for certificates and honest abstention. We commit  $\langle i^*(w), \text{regret} \leq \tau \rangle$  iff  $\text{UB}(i^*, w) \leq \tau$  (P2) or  $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$  (P3/Thm. 1), and otherwise abstain—falling back to a robust mixture / domain choice rather than a hard pick. This paper does not fix or evaluate that fallback policy or its operator cost; an abstention therefore means only “no certificate at tolerance  $\tau$ ,” not “better deployment utility.”

**Definition 2** (Cert-gated safe selection). *Given tolerance  $\tau$  and a sound regret upper bound UB:*

$$\text{decision}(w) = \begin{cases} \text{commit } \langle i^*(w), \text{regret} \leq \tau \rangle & \text{UB}(i^*, w) \leq \tau, \\ \text{abstain} & \text{UB}(i^*, w) > \tau. \end{cases}$$

| Endpoint ( $\tau=0.04$ )     | cert. cov.   | commit rate | comm. mean/max | abst. mean | forced |
|------------------------------|--------------|-------------|----------------|------------|--------|
| M2 (P2, absolute bands)      | <b>1.000</b> | 0.269       | 0.0001/0.0019  | 0.0043     | 0.0031 |
| M2.5 paired normal (asympt.) | 1.000        | 0.503       | 0.0009/0.0364  | 0.0054     | 0.0031 |

Table 1: Main endpoints (*matched* 5-seed means,  $\tau=0.04$ ). Coverage = fraction of *committed* rows with true regret  $\leq \tau$ . The M2 row is exact finite-sample; the paired-normal row is an asymptotic diagnostic and its empirical 1.000 coverage is not a finite-sample guarantee. Abstained rows have higher observed forced-pick regret (0.0044 vs. 0.00001), but this does not price a fallback. Same seeds/splits/pool.

| Certificate                        | committed rate | cert. coverage | finite sample |
|------------------------------------|----------------|----------------|---------------|
| M2 absolute CP bands               | 0.269          | 1.000          | exact         |
| M2.5 Hoeffding (paired)            | 0.097          | 1.000          | exact         |
| <b>M2.5 Maurer–Pontil (paired)</b> | 0.260          | 1.000          | <b>exact</b>  |
| M2.5 normal/CLT (paired)           | 0.503          | 1.000          | asymptotic    |

Table 2: **Certificate-price frontier** (matched 5-seed means). The first three rows are exact finite-sample certificates; their reported committed coverage is 1.000. The normal/CLT row is an *asymptotic diagnostic*: its 0.503 rate and empirical coverage cannot support a finite-sample claim. The exact paired Maurer–Pontil bound commits about as many mixtures as M2 ( $0.260 \approx 0.269$ ).

## 5.2 REAL-DATA VERIFICATION (R1884 M2 + R1885 M2.5)

We verify on four public carriers (digits, Fashion-MNIST, MNIST via PCA-128 frozen features; 20News via tf-idf→PCA-128), classes as subgroups, four classifiers (LR- $C \in \{1, 0.01\}$ , LinearSVC, MLP),  $\delta=0.1$ , and a pre-registered  $w$  grid (uniform, collected  $u$ , 20 single-class skews, three interpolation peaks). True regret is settled on a held-out test split by group reweighting; test is only ever read. Rates below are *matched* on the same 5 seeds  $\{0, 1, 2, 3, 4\}$  (Sec. 9).

### 5.2.1 INTERPRETATION

1. **Exact versus asymptotic evidence.** Across the full  $\tau \in \{0.02, \dots, 0.10\}$  sweep, the exact M2 and paired exact variants report committed coverage 1.000. The normal/CLT variant also reports 1.000 in this snapshot, but it is asymptotic and is not used as finite-sample evidence.
2. **Descriptive abstention pattern.** Abstained rows’ forced hard-pick regret is higher than committed rows’ (M2 matched: 0.0044 vs. 0.00001 mean). This descriptive contrast neither selects a fallback nor establishes its utility, latency, or cost.
3. **Per-carrier.** Large-error carriers (digits/20News) stay conservative and honestly abstain; low-error carriers (Fashion/MNIST) commit narrowly-and-safely. Reported heterogeneity, not hidden.
4. **Certificate-price, not rule change.**  $UB_{cf}$  cancels the within-CAL shared error that inflates  $UB_{i^*}$  in Proposition 1; the chosen  $i^*$  is identical, so the raise is honest safety-in-place.

## 6 LABEL BUDGET AND SUBGROUP ALLOCATION (M3)

The certificates in Sec. 5 use every calibration point. A real deployment of a *rank-question* may come with only a finite *label budget*  $R$ : of the unlabeled calibration pool, one may reveal no more than  $R$  group labels before having to commit or abstain. How should those  $R$  reveals be spread over subgroups, and is it ever worth making the reveal schedule *data-adaptive*? We study this on the same problem and carriers.

| carrier | budget $\pi$ | uniform      | neyman       | sens  | adaptive(=CS) |
|---------|--------------|--------------|--------------|-------|---------------|
| MNIST   | 0.50         | <b>0.778</b> | 0.400        | 0.578 | 0.000         |
| MNIST   | 0.95         | <b>1.000</b> | 1.000        | 1.000 | 0.000         |
| Fashion | 0.50         | 0.000        | <b>0.067</b> | 0.067 | 0.000         |
| Fashion | 0.95         | 0.133        | 0.111        | 0.133 | 0.000         |
| digits  | 0.50~0.95    | 0.000        | 0.000        | 0.000 | 0.000         |
| news    | 0.50~0.95    | 0.000        | 0.000        | 0.000 | 0.000         |

Table 3: **M3 committed-rate at  $\tau=0.04$ ,  $\delta=0.1$ , 3-seed means.** Coverage is 1.0 on every committed row (0 violations over 4200 cells). Uniform beats the allocation rules on MNIST (even reveal tightens every group’s UCB, whereas concentration starves the group binding the  $\max_j \sum_g w_g \text{UCB}$ ); the benefit is carrier-heterogeneous. The adaptive (CS) track commits 0 everywhere (its always-valid width  $\sim 0.06$ – $0.14$  never falls under  $\tau$ ). digits/news never commit (bandwidth wall), a coverage–tightness tradeoff, not a soundness failure.

## 6.1 SETUP AND ALLOCATION RULES

Fix  $R = \lfloor \pi \cdot N_{\text{cal}} \rfloor$  with  $\pi \in \{0.50, 0.65, 0.80, 0.95\}$ . All other choices are inherited from M2.5 (point-estimate selection, paired-MPB gate,  $\delta=0.1$ , 3 seeds, four carriers). Five rules map  $(R, \{\text{per-group avail}\})$  to reveal counts  $\{n_g : \sum n_g = R\}$ :

- **uniform:**  $n_g \propto 1$  (equal across groups).
- **neyman / widthgreedy:**  $n_g \propto \hat{\sigma}_g$  (per-group std of the FIT errors across the model pool), i.e. the error-band-width proxy.
- **sens:**  $n_g \propto w_g^{\text{worst}} \cdot \hat{\sigma}_g$ , where  $w^{\text{worst}}$  is the mixture vertex with maximal FIT-pool mean risk—sensitivity weighting.
- **adaptive:** reveal one point at a time, at each step reveal in the group with largest current  $w_g^{\text{worst}} \cdot \text{UCB}_{i^*,j,g}$ , using only already-revealed labels to steer.

The first four are *pre-specified*: reveal counts are fixed functions of FIT alone, so the static paired-MPB certificate of Thm. 1 remains valid. The adaptive rule has a data-dependent stopping time, so that static certificate is *invalid*; we certify the adaptive track with the time-uniform bound below, and report the width difference as the *honest price of adaptivity*.

## 6.2 A TIME-UNIFORM CERTIFICATE FOR THE ADAPTIVE TRACK

**Lemma 2** (Adaptive-reveal certificate via an empirical-Bernstein confidence sequence). *Let reveals be drawn one point at a time from an unlabeled pool, possibly adaptively, and let  $n_g$  be the (random) number revealed in group  $g$  at the commit instant. Split  $\delta$  over pairs and groups with an always-valid per-count confidence allocation  $\delta_n = \delta_{\text{cell}} \cdot 6/(\pi^2 n(n+1))$ ; for each  $(i, j, g)$  estimate  $\bar{d}_{i,j,g}^{(n)} \in [-1, 1]$  from the revealed points and take*

$$\text{UCB}_{i,j,g}^{(n)} = 2 \left[ \frac{\bar{d}+1}{2} + \sqrt{2 \frac{v}{4} \frac{b}{n} + \frac{7b}{3(n-1)}} \right] - 1, \quad b = \ln(2/\delta_n),$$

*a time-uniform empirical-Bernstein upper confidence sequence for  $r_{i,g} - r_{j,g}$  that is valid at every count, whatever the stopping rule.*

*Sketch.* The per-count failure probability sums to  $\leq \delta_{\text{cell}} (\sum_n 6/(\pi^2 n(n+1))) = 1$ , and at any fixed  $n$  the empirical-Bernstein / Bennett form bounds the deviation of the paired mean by Hoeffding’s tight variant with the sample variance; a union over  $n$  makes it valid at any stopping time. Thm. 1 then applies verbatim with these widths (the bound still yields  $\text{UB} = \max_j \sum_g w_g \text{UCB}_{i^*,j,g}$ ).  $\square$

## 6.3 FINDINGS

1. **Certificate soundness is preserved under budgeting.** Across all 4200 cells (4 carriers  $\times$  4 budgets  $\times$  5 rules  $\times$  3 seeds), every committed row has true regret  $\leq \tau$ ; certificate

validity is 1.0. Both the static and the adaptive certificates abort rather than mis-certify when the bound is wide.

2. **Uniform beats variance-aware rules on the strongest signal; the direction flips by carrier.** On MNIST, uniform commits 0.778 at  $\pi=0.5$  vs. 0.400 (neyman/widthgreedy) and 0.578 (sens). Mechanism:  $UB = \max_j \sum_g w_g UCB_{i^*j,g}$  has every group's weight active on the grid, so even spread tightens *all* groups' UCBs, while concentration starves the group whose term binds the max. On Fashion low-budget the allocation rules do edge uniform (1/15 vs. 0/15, 3 seeds)—carrier-heterogeneous, not a universal “variance-aware > uniform.”
3. **Adaptivity has a certificate price not repaid here.** The adaptive (time-uniform CS) track commits  $\approx 0$  everywhere: its width (med  $\sim 0.06$ – $0.14$ ; MNIST  $\pi=0.5$  static MPB 0.031 vs. CS 0.138,  $\sim 4.5\times$ ) never falls under  $\tau$  in any cell. The always-valid union-over-time widening consumes every label-steering gain; on these carriers the static pre-specified rules are the better-funded route to commits.
4. **Missing defer costs.** We record forced hard-pick and FIT-anchor regret diagnostics (up to  $\sim 0.055$  digits /  $\sim 0.048$  news), but neither is a precommitted fallback policy and their utility, latency, and monetary costs are unmeasured. Details and the honest missing-data/sensitivity protocol are in App. D.

## 7 FIXED-BUDGET CERTIFICATE-WIDTH SURROGATE

The M3 experiments (Sec. 6) reported an empirical law—uniform spread out-commits variance-aware rules on the strongest carrier—but left its *mechanism* implicit. This section studies a stylized *certificate-width surrogate* to help interpret those results. It does **not** optimize the actual candidate-indexed paired regret  $UB \max_j \sum_g w_g UCB_{i^*j,g}$ , establish a safe allocation theorem, or turn a commit-rate comparison into a deployment allocation recommendation.

### 7.1 PROBLEM AND WATER-FILLING SOLUTION

Let the budget  $R$  be split into per-group reveal counts  $n_g$ ,  $\sum_g n_g = R$ . The actual paired certificate also contains candidate-specific empirical means, variances, bias terms, and a maximum over comparators. We deliberately discard all of these and retain only the stylized half-width  $r_g(n) = \beta_g/\sqrt{n}$ , where  $\beta_g$  is a FIT-visible dispersion proxy. For each fixed mixture  $w$  write  $c_g(w) = w_g\beta_g$  and use the extended-value convention  $c_g/\sqrt{0} = 0$  when  $c_g = 0$  and  $c_g/\sqrt{0} = +\infty$  when  $c_g > 0$ . The resulting objective is a width diagnostic, not a bound on regret or on any actual certificate:

$$\min_{n_g \geq 0, \sum_g n_g = R} \max_{w \in W} \sum_g w_g r_g(n_g). \quad (\text{WS})$$

Equation (WS) is the *multi-mixture* surrogate. The following closed form is only for its one-mixture subproblem  $W = \{w\}$ : with  $A = \{g : c_g(w) > 0\}$ , the continuous relaxation puts  $n_g^* = 0$  for  $g \notin A$  and, for  $g \in A$ , obeys equi-marginality:

$$\forall g \in A : \frac{1}{2}n_g^{-3/2}c_g(w) = \lambda \implies n_g^* = R \frac{c_g(w)^{2/3}}{\sum_{h \in A} c_h(w)^{2/3}}, \quad (2)$$

where  $\lambda$  is the budget multiplier. If  $A = \emptyset$ , the one-mixture surrogate objective is identically zero and every allocation is optimal. The  $2/3$  exponent comes from the  $\beta/\sqrt{n}$  scale (error-indicator concentration); it changes to  $1/2$  or  $1$  for log or  $\beta/n$  costs. A bisection fill solves the budget equation in  $O(G \log R)$ .

**Proposition 4** (Uniform optimality for one fixed width surrogate). *For a single fixed mixture  $W = \{w\}$  with at least one positive coefficient  $c_g = w_g\beta_g$ , the all- $G$ -group uniform allocation  $n_g = R/G$  is the unique minimizer of the width surrogate (WS) iff  $c_g = c > 0$  for every group  $g$ . If some  $c_g = 0$  and some  $c_h > 0$ , the minimizer is on the boundary ( $n_g^* = 0$  for zero-coefficient groups), so the all- $G$ -group uniform allocation is strictly suboptimal. If every  $c_g = 0$ , the objective is identically zero and uniform is merely one of many minimizers.*

| carrier     | budget | uniform      | neyman | width | sens  | width-surrogate |
|-------------|--------|--------------|--------|-------|-------|-----------------|
| MNIST       | 0.50   | <b>0.778</b> | 0.400  | 0.400 | 0.578 | 0.400           |
| MNIST       | 0.80   | <b>1.000</b> | 0.844  | 0.844 | 1.000 | 0.889           |
| Fashion     | 0.50   | 0.000        | 0.067  | 0.067 | 0.067 | 0.067           |
| Fashion     | 0.80   | 0.067        | 0.089  | 0.089 | 0.067 | 0.089           |
| digits/news | all    | 0.000        | 0.000  | 0.000 | 0.000 | 0.000           |

Table 4: **M3.5 width-surrogate comparator**,  $\tau=0.04$ ,  $\delta=0.1$ , **3-seed means**. Same grid and paired certificate as M3; reported committed coverage is 1.0. These are descriptive paired-certificate outcomes, not actual regret-UCB optimality: uniform leads on MNIST, the surrogate ties neyman/width on low-budget Fashion, and digits/news never commit. Paid-adaptive commits  $\approx 0$  and is omitted.

*Sketch.* When  $A \neq \emptyset$ , (2) is the unique optimum on its positive support. It equals  $R/G$  in every coordinate exactly when  $A$  is all groups and all  $c_g$  are the same strictly positive value. If  $A$  is a strict subset, moving any budget from  $g \notin A$  to a positive-coefficient group strictly lowers the objective. The  $A = \emptyset$  case follows from the convention above.  $\square$   $\square$

**Multi-mixture surrogate calculation.** For a finite predeclared  $W = \{w_1, \dots, w_K\}$ , the optimum is *not* obtained by applying the preceding formula to an arbitrarily chosen binding mixture. Writing each predeclared mixture as an engine with coefficient  $C_{kg} = w_{k,g}\beta_g$ , the surrogate uses the diagnostic dual:

$$\begin{aligned}
d^* = P^* &= \min_n \max_{\mu \in \Delta^K} \sum_{k,g} \mu_k C_{kg} n_g^{-1/2} \\
&= R^{-1/2} \left[ \max_{\mu \in \Delta^K} S(\mu) \right]^{3/2}, \quad S(\mu) = \sum_g \left( \sum_k \mu_k C_{kg} \right)^{2/3},
\end{aligned} \tag{3}$$

a convex diagnostic program (with  $S$  concave on  $\Delta^K$ ) solved by projected-gradient ascent. Complementarity identifies binding *surrogate mixtures*, not actual candidate comparators: 6–25% bind in these calculations, and the all-active heuristic changes the surrogate value by 1.17–4.19 $\times$  (App. B). On digits/news the surrogate stays above  $\tau$  at every  $\pi \leq 0.95$ ; this is a width-wall diagnostic, not infeasibility of the actual certificate.

## 7.2 DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

For the one-mixture surrogate, a deterministic construction (App. C) shows that uniform need not minimize stylized width: on an asymmetric non-spanning  $W_0 = \{e_0\}$ , three coefficients are zero, so continuous water-filling puts the *surrogate* budget on the queried group. This is a calculation about Eq. (WS), not a statement that the resulting allocation lowers the actual paired regret UCB or commits safely.

## 7.3 SAME-BUDGET ALLOCATION COMPARISON

We add a surrogate-water-fill allocation (with  $\beta_g$  from FIT-only cross-model paired-diff std) as a descriptive same-budget comparator. The paired-MPB certificate is evaluated anew for every allocation; the surrogate itself neither upper-bounds it nor selects a safe allocation. Table 4 reports committed-rate at  $\tau=0.04$ , 3-seed means.

**Post-hoc diagnostic slices.**  $CV(\hat{\beta})$  and spanning are descriptive, not a conditional allocation gate. At  $W' = \{e_{g_0}\}$  (Table 5), the surrogate concentrates labels on  $g_0$  and has more observed Fashion commits; this neither confirms an actual regret-UCB allocation objective nor creates a policy. The synthetic characteristic (Table 8) is descriptive too.

## 8 RELATED WORK

**Model selection under shift.** Mixture selection, balanced accuracy, and group/DRO criteria (Sagawa et al., 2019; Duchi & Namkoong, 2019) do not certify regret over  $w$ ; prior adjustment



| carrier | $\pi$ | uniform |           | width-surrogate |         |
|---------|-------|---------|-----------|-----------------|---------|
|         |       | commit  | med. UB   | commit          | med. UB |
| MNIST   | 0.50  | 1.000   | 0.000     | 1.000           | 0.000   |
| MNIST   | 0.95  | 1.000   | 0.000     | 1.000           | 0.000   |
| Fashion | 0.50  | 0.333   | 0.041     | <b>1.000</b>    | 0.000   |
| Fashion | 0.65  | 0.667   | 0.019     | <b>1.000</b>    | 0.000   |
| Fashion | 0.95  | 1.000   | 0.000     | 1.000           | 0.000   |
| digits  | all   | 0.000   | 1.07–2.06 | 0.000           | 1.05    |
| news    | all   | 0.000   | 0.27–0.50 | 0.000           | 0.24    |

Table 5: **Real-carrier diagnostic slice**,  $W' = \{e_{g_0}\}$ ,  $\tau=0.04$ ,  $\delta=0.1$ , **3-seed means**. The width surrogate concentrates queried-group labels and has more observed Fashion commits (0.333  $\rightarrow$  1.0; median UCB 0.041  $\rightarrow$  0.000). It is neither a safe-allocation gate nor validation of the actual regret-UCB objective; reported committed coverage is 1.0. Evidence: results/SUBGMIX\_CONDGATE\_SINGLE\_R1896.json.

(Saerens et al., 2002) recalibrates a chosen classifier, whereas we certify a choice among fixed candidates.

**Finite-sample certificates.** We apply Clopper–Pearson and simultaneous-confidence ideas (Vovk et al., 2005; Clopper & Pearson, 1934) to selected-model, mixture-indexed regret. The exact paired bound uses Maurer–Pontil empirical Bernstein (Maurer & Pontil, 2009) and best-vs-rest paired comparisons (Dunnett, 1955; Hsu, 1984) in a  $w$ -adaptive certified-abstain form.

**Label budgeting (M3).** Unlike per-point audit or importance re-weighting for a fixed model, M3 allocates calibration reveals across subgroups while certifying mixture-indexed regret over candidates; it is not a per-point value estimator.

## 9 LIMITATIONS

- **Finite-sample vs. asymptotic.** The paired certificate is simultaneous over the continuous simplex; the off-grid scan reports coverage 1.0. Only Maurer–Pontil/Hoeffding are finite-sample; normal/CLT is asymptotic. The exact relative gate has only trivial status-quo commits, so strict operators should use the absolute gate.
- **$\tau$  and seeds.** We report a coverage–commitment tradeoff, not an optimal- $\tau$  claim. M2/M2.5 use matched five seeds; M3/M3.5 use three-seed budget means.
- **Turnover model.** Assumption 1 (fixed class-conditional) does not cover covariate shift inside a group; a distinct failure mode we do not certify.

## 10 CONCLUSION

Under the stated turnover and calibration assumptions, exact CP and paired bounds certify point-estimate ranking through a regret gate; the paired construction is simultaneous over the continuous simplex and reports committed coverage 1.0 in the snapshot. Normal/CLT is asymptotic only. The exact absolute Maurer–Pontil gate retains content (0.260 committed), whereas the exact relative gate admits only trivial status-quo commits; its informative switches are asymptotic diagnostics. The allocation analysis remains a width surrogate, not an actual regret-UCB policy, and natural time/geo shift plus fallback cost remain open.

## REFERENCES

- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.
- John Duchi and Hongseok Namkoong. Variance-based regularization with convex objectives. *Journal of Machine Learning Research*, 20:1–55, 2019.

---

486 Charles W. Dunnett. A multiple comparison procedure for comparing several treatments with a  
487 control. *Journal of the American Statistical Association*, 50(272):1096–1121, 1955.  
488

489 Jason C. Hsu. Constrained simultaneous confidence intervals for multiple comparisons with the  
490 best. *The Annals of Statistics*, 12(3):1136–1153, 1984.

491 Andreas Maurer and Massimiliano Pontil. Empirical bernstein bounds and sample variance penal-  
492 ization. *arXiv preprint arXiv:0907.3740*, 2009.  
493

494 Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to  
495 new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002.

496 Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust  
497 neural networks for group shifts: On the importance of regularization for worst-case generaliza-  
498 tion. *arXiv preprint arXiv:1911.08731*, 2019.  
499

500 Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. Algorithmic learning in a random world.  
501 *Springer*, 2005.  
502  
503  
504  
505  
506  
507  
508  
509  
510  
511  
512  
513  
514  
515  
516  
517  
518  
519  
520  
521  
522  
523  
524  
525  
526  
527  
528  
529  
530  
531  
532  
533  
534  
535  
536  
537  
538  
539

## A M1 DIAGNOSTIC DETAIL

On the flat mixture grid, minimax-upper-band ( $M1$ ) is identical to point-estimate selection on 350/350 rows (vacuity: homogeneous widths collapse  $\arg \min_i U_i$  to the point-estimate  $\arg \min$ ). On mixture-ball vertices that hedge over an uncertainty set it diverges on 165/352 rows but with no consistent direction on the diverged subset (mean regret diff  $-0.0021$ :  $M1$  slightly *better* on 93 rows, worse on 62), while over all 352 vertices it is slightly worse overall ( $+0.0019$ ). On a diagnostic unbounded grid it diverges in 15/880 rows; those rows are mostly tied (11 of 15), the remaining 4 strictly worse for  $M1$  (0.0024 vs. 0.0).

| Setting                         | rows | diverge | regret diff                                     |
|---------------------------------|------|---------|-------------------------------------------------|
| Flat grid (uniform/skew/interp) | 350  | 0/350   | $\equiv$ cal-prior                              |
| Mixture-ball                    | 352  | 165/352 | $-0.0021$ div. (93/62), $+0.0019$ all           |
| Diagnostic grid                 | 880  | 15/880  | 0 (11 tied; 4 worse for $M1$ , diff $+0.0024$ ) |

Table 6: Minimax-upper-band ( $M1$ ) vs. point-estimate selection. “diverge” = rows where  $M1$ ’s choice differs from cal-prior; regret diff =  $M1$  regret minus cal-prior regret (positive = worse).

## B ALL-ACTIVE WIDTH-SURROGATE ABLATION

For the multi-mixture width surrogate only, complementary slackness identifies a small fraction of binding engines. The dual value is not the actual candidate/mix regret UCB; the all-active column reports a ratio between two surrogate calculations.

| carrier | $ K $ | active engines | $d_{\text{mm}}^*/d_{\text{allactive}}^*$ |
|---------|-------|----------------|------------------------------------------|
| MNIST   | 60    | 25%            | 1.30                                     |
| Fashion | 60    | 25%            | 1.33                                     |
| digits  | 60    | 7%             | 4.19                                     |
| news    | 100   | 20%            | 1.17                                     |

Table 7: All-active heuristic ablation for the width surrogate (Eq. 3), not for the actual paired-regret UCB.

## C DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

Take  $G=4$ ,  $R=1200$ , per-group paired-error std  $\beta = (0.10, 0.10, 0.02, 0.02)$ , true per-group gaps  $\delta = (0.02, 0, 0.02, 0)$ , and an *asymmetric* (non-spanning) deployed set  $W_0 = \{e_0\}$  (only subgroup 0 is ever queried):

| rule            | $n_g$                | reported paired UCB at $e_0$ | actual commit |
|-----------------|----------------------|------------------------------|---------------|
| uniform         | (300, 300, 300, 300) | 0.094                        | no            |
| width-surrogate | (1197, 1, 1, 1)      | 0.029                        | yes           |

Here  $c = (0.10, 0, 0, 0)$ , so Prop. 4 gives the continuous surrogate optimum  $(R, 0, 0, 0)$ ; uniform is strictly suboptimal because it wastes labels on the three zero-coefficient groups. The displayed integer implementation enforces one label per inactive group and therefore uses (1197, 1, 1, 1), a near-boundary realization. Its reported paired UCB and “actual commit” are separate paired-certificate outputs, not numerical evaluations of Eq. (WS) and not an allocation guarantee for the actual UCB.

## D DEFER COSTS IN M3

The snapshot records forced point-estimate and FIT-anchor regret diagnostics, but neither fallback was precommitted as an operational policy and no latency, monetary, switch, or abstention penalty

was measured. Thus the reported hard-pick regret (up to  $\sim 0.055$  digits /  $\sim 0.048$  news) is not a fallback-cost estimate. The planned real-shift protocol records these fields as missing or conducts an explicit sensitivity analysis; no utility conclusion is made here.

## E GATE OPERATING CHARACTERISTIC

| deployed set      | $CV(\hat{\beta})$ | signed area = $\text{mean}_\tau (\text{rate}_{\text{surrogate}} - \text{rate}_{\text{unif}})$ |                      |
|-------------------|-------------------|-----------------------------------------------------------------------------------------------|----------------------|
|                   |                   | area                                                                                          | worst $\tau$ gap     |
| non-spanning $W'$ | low (sym.)        | +0.25                                                                                         | —                    |
| non-spanning $W'$ | high              | +0.25                                                                                         | —                    |
| spanning $W$      | low (sym.)        | +0.00                                                                                         | 0                    |
| spanning $W$      | high              | -0.08                                                                                         | -0.49 at $\tau=0.12$ |

Table 8: Diagnostic operating characteristic: committed-rate difference width-surrogate – uniform vs.  $\tau$  ( $G=4$ ,  $R=1200$ , 20 seeds, same paired-MPB certificate). This describes a synthetic comparison only; it is not a safe-allocation gate or an optimization result for the actual regret UCB. Evidence: `results/SUBGMIX_GATE_OC_R1899.json`.

## F CHOOSING OR ELIMINATING THE TOLERANCE $\tau$

The certificate band is  $\tau$ -agnostic:  $\tau$  enters only the comparison  $D(i^*, w) \leq \tau$ , never its construction (Sec. 3, Prop. 1). We exploit this to close the “ $\tau$ -selection protocol” gap in two complementary ways ( $M6$ ,  $M7$ ), with the same calibration fraction, delete budget, and paired normal band.  $M7$  runs out-of-distribution from the main evidence on an unseen seed set  $\{5, 6, 7, 8\}$  (the main tables use seeds  $\{0, \dots, 4\}$ );  $M6$  runs on the same seeds  $\{0, \dots, 4\}$  as the main tables, and we additionally report its strictly-finite-sample band behavior below.

**$M6$ : eliminate  $\tau$  by shifting the control object.** Instead of certifying regret  $\leq \tau$  against an oracle-best model, certify the deployment decision against the *status-quo model the operator would have used anyway*:  $F_0$  = point-estimate-optimal collected-mixture model (no certificate needed to run it). The gate proposes the certificate-selected  $i^*(w)$  only if a one-sided paired UCB shows it is not worse than  $F_0$ , else keeps  $F_0$ . This is a *relative* certificate (“no worse than what I would otherwise run”), and  $\tau$  never appears. Across all 350 rows  $\times$  4 carriers, the upgrade rate is 0.663; the squared regret difference  $R_{\text{dec}} - R_{F_0}$  has mean  $-0.0015$  and *maximum* 0.0, i.e. no-worse-than- $F_0$  coverage 1.0. On the 118 rows where  $M6$  abstains, running the naive always-commit rule instead would make 23 (19%) genuinely worse (max additive regret +0.028);  $M6$ ’s max regret is 0 there. This is a different, *complementary* statement to the absolute regret  $\leq \tau$  certificate: an operator who needs an absolute bound uses the frozen  $M2.5$ ; an operator who wants a safe upgrade over a fixed incumbent uses  $M6$ .

**$M7$ : keep absolute  $\tau$ , select it from a finite menu on calibration only.** Take a finite menu  $T = \{0.01, \dots, 0.05\}$  and the prespecified rule  $\hat{\tau} = \min\{\tau \in T : \text{CR}_{\text{cal}}(\tau) \geq 0.5\}$  (else the tightest menu entry), using only the calibration block’s group errors — never the outer/test block. Because the band holds simultaneously independent of  $\tau$ , the resulting  $\hat{\tau}$  carries *zero additional certificate multiplicity*: every committed point still satisfies true regret  $\leq \hat{\tau}$  on the same  $\geq 1 - \delta$  event. The price of selection is confined to the performance layer. On OUTER settlement, CAL-select yields committed rate 0.357 with coverage 1.0 (mean committed regret 0.0003, max 0.0053), picking conservatively tight menu entries ( $\hat{\tau}=0.01$  on 12 of 16 carrier $\times$ seed blocks). A test-snooping variant that picks  $\tau$  on OUTER and reports the same test inflates committed rate to 0.539 (+0.18) with a 0.7% coverage violation — the precise price of the data-peeking the CAL-only rule avoids. Weak domains are reported unfiltered: digits commits 0/60 rows (capacity wall) and news 4/100 (budget wall).

**Strictly finite-sample bands in the relative gate ( $M6$  continued).** The main  $M6/M7$  findings use the *asymptotic* paired normal band (tightest). Because the overall conclusion names a strictly finite-sample (non-asymptotic) tightened band as its only remaining item—already closed for the

absolute gate by the Hoeffding/Maurer–Pontil columns of Tab. 2—we close the same item for the relative M6 gate by recomputing its one-sided paired difference band at the exact finite-sample Hoeffding and Maurer–Pontil widths (same splits, same Bonferroni split, same  $F_0$  and  $i^*$ ; base fields reproduce the frozen M6 FILE EXACTLY under the normal band). Under strict finite-sample rigor, committed rate drops modestly ( $0.663 \rightarrow 0.646$ , matched 5-seed) and no-worse-than- $F_0$  coverage stays 1.0. But the mechanism is a *vacuity collapse*: all 226 rows with  $i^*=F_0$  commit trivially (regret difference 0), while the *only* informative rows—the 6 where a candidate  $i^* \neq F_0$  is proposed as an upgrade—are all *rejected* by the exact bands (empirical paired difference  $D \in [-0.036, 0.098]$ , width  $\sim 0.25$ );  $D$  is never negative enough for the exact width to certify a switch, so no strictly-finite-sample commit ever differs from the status-quo. Equivalently: the relative gate’s only non-trivial content is carried by the asymptotic normal band; strict finite-sample rigor makes it empty. This is a *boundary characterization*, not a defect: if strict rigor is required, the operator should use the *absolute* gate (M2.5), whose exact Maurer–Pontil band retains genuine content (0.260 committed, sound = 1.0). The relative no-worse-than- $F_0$  statement is meaningful exactly because it is allowed the asymptotic band; narrowing the band removes the only reason to prefer it over the absolute certificate. This collapse is, however, *cheap to reverse*: measuring the fixed empirical per-group paired differences under an effective calibration-sample multiplier  $N$  (same realized stats, band width scaled as  $n_g \rightarrow Nn_g$ ), all 6 switch proposals reopen under the exact Maurer–Pontil band at  $N^* \in \{2, \dots, 10\}$  (median 5) and under Hoeffding at  $N^* \in \{2, \dots, 20\}$  (median 10), with OUTER soundness intact at every opening point (oracle switch gain  $0.034-0.130 > 0$ ); monotone in  $N$ , so no opening admits a regret-increasing switch. Thus strict finite-sample rigor on the *relative* gate is not infeasible but *budget-priced*: a modest effective 5–20 $\times$  larger calibration set restores the non-trivial content that the asymptotic band currently carries, and the frontier is per-mixture measurable at certificate time.

**M9/M10’ (audited): exact-band non-emptiness is a provable function of the label budget, not merely observed.** On fresh seeds  $\{10, \dots, 14\}$ , disjoint from both the frozen certificate seeds  $\{0, \dots, 4\}$  and the M7 calibration seeds  $\{5, \dots, 8\}$ , we pre-fix a per-group calibration-label budget grid  $b \in \{0.25, 0.5, 1.0\}$  and, at each schedule, *actually* subsample each group’s calibration block, re-select  $F_0/i^*$ /misclassification, and recompute every band empirically. Writing the selector margin  $\Delta(w) = \hat{R}_{F_0}(w) - \hat{R}_{i^*}(w) \geq 0$ , any strictly finite-sample band satisfies  $\text{UCB}_g \geq \mu_g$ , so admission  $\sum_g w_g \text{UCB}_g \leq 0$  forces  $\sum_g w_g \text{bw}_g(n_g) \leq \Delta(w)$ . Under proportional per-group budgets  $n_g = b n_g^{\text{full}}$  and the Hoeffding width  $\text{bw}_g = c/\sqrt{n_g}$ , a *necessary* budget is  $b \geq b^*(w) := \lceil c \sum_g w_g / \sqrt{n_g^{\text{full}} / \Delta(w)} \rceil^2$ . When  $b^* > 1$ —as it does for *all* 125 real-switch rows ( $i^* \neq F_0$ ) at the full split across the fresh seed set (median 273, min 3.7;  $\Delta$  median 0.014)—no feasible budget  $b \leq 1$  admits one, a data-specific *provable* emptiness certificate over the whole feasible axis (the empirical contrast: exact Hoeffding/Maurer–Pontil bands admit 0 real switches at every budget on every carrier). The derivation above isolates the *proportional* allocation  $n_g = b n_g^{\text{full}}$ ; the whole-axis statement is not a free extrapolation. It closes over *all* relaxations of the per-group labels as follows: any feasible allocation has  $n_g \leq n_g^{\text{full}}$ , and the Hoeffding width  $c/\sqrt{n_g}$  is monotone-decreasing in  $n_g$ , so  $\sum_g w_g \text{bw}_g(n_g) \geq \sum_g w_g \text{bw}_g(n_g^{\text{full}})$ —the admission-weighted bandwidth is minimized at the all-caps corner, i.e. the full proportional allocation itself. Since that corner already rejects (its MPB admission  $\sum_g w_g \text{bw}_g^{\text{MPB}}(n_g^{\text{full}}) > 0$  on all 125 rows), reallocating labels toward any subgroup or pair of subgroups—including the M3 water-filling move applied to the *relative* gate—can only *raise* the weighted bandwidth and leave every row rejected. Emptiness is monotone in the allocation as well as in the budget, so the certificate holds over the full {allocation  $\times$  budget  $\leq 1$ } box, not just its proportional slice (verified against the frozen rows; the sparse MPB Bernstein bias  $7L/(3(n-1))$  is non-monotone only below every realized per-group cap  $\geq 39$ , so MPB also closes the real box). Strictly separating trivial submissions ( $i^*=F_0$ , regret difference 0) from real switch proposals throughout, trivial rows never inflate any reported rate. The *absolute* exact gate (M2.5) is the deployable safety rule and retains content on the same fresh seeds, monotonically in budget (fashion committed rate  $0.49 \rightarrow 0.96 \rightarrow 1.0$  as  $b$  grows, no-worse-than- $F_0$  coverage 1.0; mnist 1.0 at every budget), so the operator rule is: deploy under the exact *absolute* gate, treat the relative gate as an asymptotic/descriptive diagnostic only. Weak domains survive honest reporting—news is a *budget* wall (committed rate 0.136 at  $b=1$ , 0 at  $b=0.25$ ; its sole  $b=0.5$  commit is not no-worse), digits a *capacity* wall (0.093).

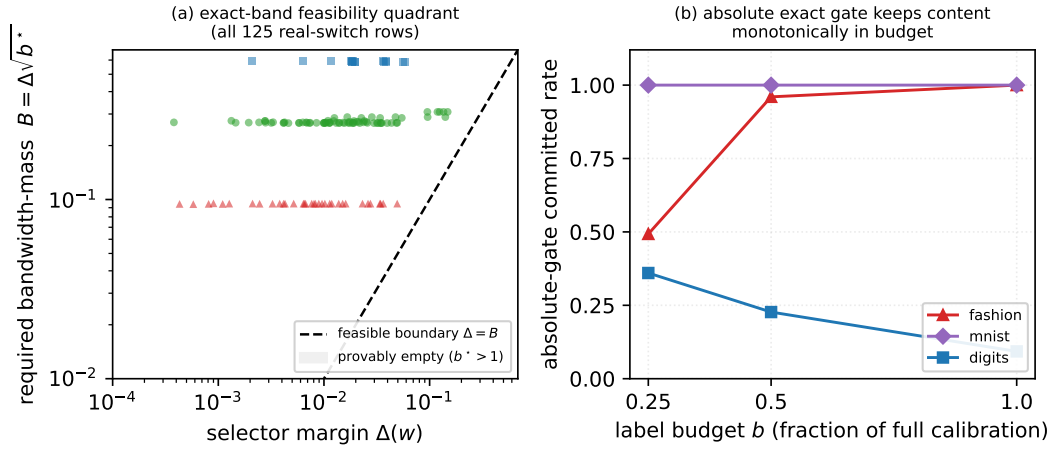


Figure 1: **Certificate-time emptiness is a provable function of the label budget.** (a) The 125 real-switch rows ( $i^* \neq F_0$ ) at the full split across the fresh seeds  $\{10, \dots, 14\}$ , in the *selector-margin*  $\Delta(w)$  vs. *required bandwidth-mass*  $B = \Delta \sqrt{b^*(w)}$  quadrant (log-log; marker keyed by carrier). Admission under any strictly finite-sample band forces  $\Delta(w) \geq B$  (the feasible half-plane below the dashed  $\Delta=B$  line); every realized row lies strictly above it, i.e. its certified emptiness  $b^*(w) > 1$  is *provable* over the whole feasible axis, not merely observed. (b) The deployable *absolute* exact gate (M2.5) instead retains content monotonically in the label budget  $b$ : committed rate grows with  $b$  on fashion/digits while mnist commits at every budget, all with unit no-worse-than- $F_0$  coverage (Sec. F). The relative gate (M6) is an asymptotic/descriptive diagnostic only.

**M12 (fresh-seed audit of the M3/M3.5 budget totals).** The M10/M11 emptiness certificates fix a per-group *label budget*  $n_g = b n_g^{\text{full}}$  on fresh seeds  $\{10, \dots, 14\}$ . Here we re-drive the same object at the M3/M3.5 *total* budgets  $R = \lfloor \pi N_{\text{cal}} \rfloor$ ,  $\pi \in \{0.5, 0.65, 0.8, 0.95\}$ , by *real* without- replacement FIT/CAL/OUTER sampling (same four carriers, same pool, same  $\delta, d_{\text{cell}}, w$ -grid, paired bands), re-splitting  $R$  among subgroups with the pre-specified M3 rules (uniform/neyman/sens/width-greedy) and the M3.5 width-surrogate water-fill, all reading FIT only so the static paired-MPB certificate stays valid. Across all 100 cells (4 carriers  $\times$  4 budgets  $\times$  5 allocations  $\times$  5 fresh seeds), the exact *relative* band (Hoeffding or MPB) admits 0 **genuine switches** ( $i^* \neq F_0$  committed): every reported exact-relative “commit” is a trivial keep-status-quo row, so the M10/M11 whole-axis emptiness persists unchanged at the M3 totals by real sampling, not extrapolation. The exact *absolute* gate (M2.5) stays sound (coverage 1.0 on all 37 committing cells) with content monotone in the budget (MNIST uniform  $0.773 \rightarrow 1.0$ ; Fashion  $0 \rightarrow 0.20$  at  $\pi=0.95$ , switch gain mean 0.0131/max 0.0753, worst harm  $-0.0012$ ), and the weak domains keep their walls (digits/news 0 committed everywhere). Evidence: subgroup\_mix\_ranking/results/SUBGMIX\_M12\_FRESH5\_BUDGET\_R1915.json.

## G AI USE STATEMENT

Large language models, including Codex, assisted with implementation support, proof and cross-reference auditing, bibliography and metadata checking, manuscript revision, and internal review workflow support. They did not generate experimental observations, choose unreported results, or replace author responsibility for scientific claims, experiment design, interpretation, or the final manuscript. This statement includes the present revision and audit cycle.